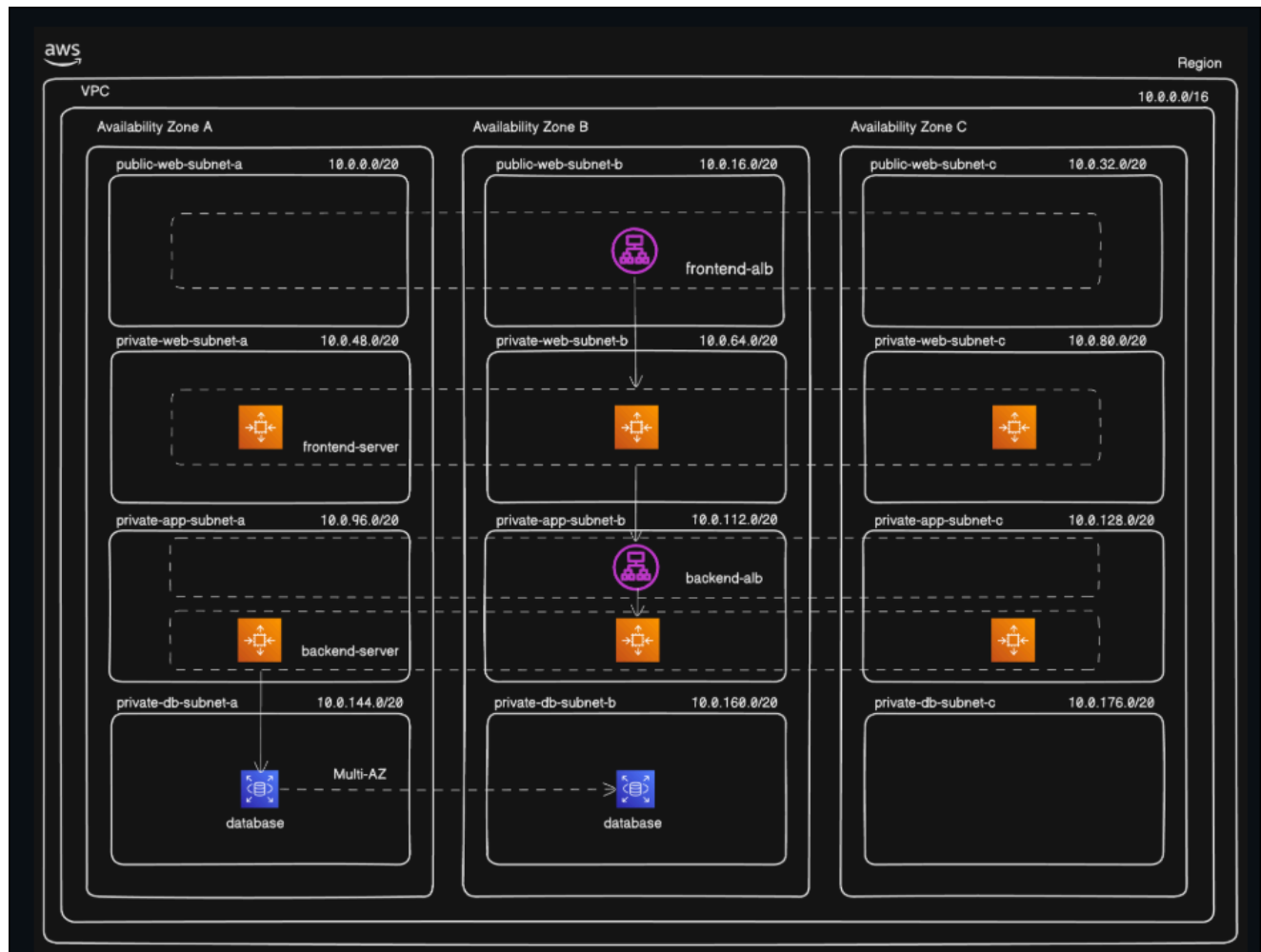


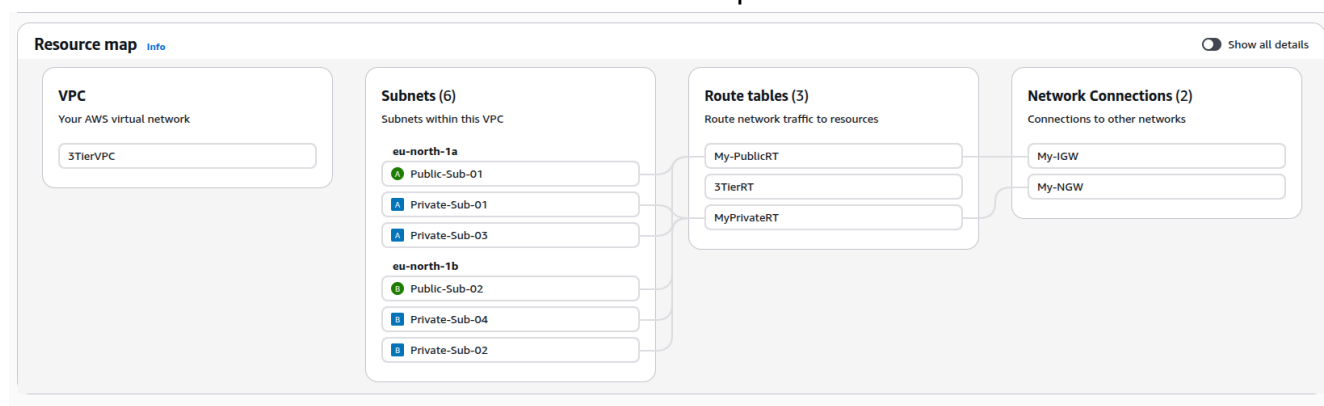
**Q1. Deploy a three-tier application as it was done in today's session, following the GitHub instructions, by tightening all security groups,restricting inbound rules only for now.**

<https://github.com/akashsawan1/three-tier-architecture-aws>



**Ans:**

i. The first step in the creation was to creation of VPC and subnets . I choose to create 2 public subnets and 4 private subnets , created an internet gateway and attached it to the public subnets and then created the nat gateway and attached it to the private subnets . The VPC resulted is shown below from the resource map of the VPC section.



ii . I created route tables for public and private subnets and associated them with the respective subnets

**rtb-075cff0dd9e293f10 / My-PublicRT** Actions

**Details** info

**Route table ID**  
rtb-075cff0dd9e293f10

**VPC**  
vpc-026632d2d69928fb0 | 3TierVPC

**Main**  
No

**Owner ID**  
596059883112

**Explicit subnet associations**  
2 subnets

**Edge associations**  
-

Routes | Subnet associations | Edge associations | Route propagation | Tags

**Routes (2)** Both Edit routes

| Destination | Target                | Status | Propagated | Route Origin       |
|-------------|-----------------------|--------|------------|--------------------|
| 0.0.0.0/0   | igw-0b8b72c09743c8a95 | Active | No         | Create Route       |
| 10.0.0.0/16 | local                 | Active | No         | Create Route Table |

**rtb-0c20c5ab48a1439bf / MyPrivateRT** Actions

**Details** info

**Route table ID**  
rtb-0c20c5ab48a1439bf

**VPC**  
vpc-026632d2d69928fb0 | 3TierVPC

**Main**  
No

**Owner ID**  
596059883112

**Explicit subnet associations**  
4 subnets

**Edge associations**  
-

Routes | Subnet associations | Edge associations | Route propagation | Tags**Routes (2)** Both Edit routes

| Destination | Target                | Status | Propagated | Route Origin       |
|-------------|-----------------------|--------|------------|--------------------|
| 0.0.0.0/0   | nat-01a8acbbd87d4a8d3 | Active | No         | Create Route       |
| 10.0.0.0/16 | local                 | Active | No         | Create Route Table |

iii. As a next step I created the security groups for frontend-alb , frontend-asg , backend-alb , backend-asg , db-sg.

**Security Groups (5)** Info Actions Export security groups to CSV Create security group

| <input type="checkbox"/> | Name | Security group ID     | Security group name | VPC ID                | Description                | Owner        |
|--------------------------|------|-----------------------|---------------------|-----------------------|----------------------------|--------------|
| <input type="checkbox"/> | -    | sg-0a644300e0dcada7b  | backend-asg-sg      | vpc-026632d2d69928fb0 | allow                      | 596059883112 |
| <input type="checkbox"/> | -    | sg-0376ad0931e59a2b6  | db-sg               | vpc-026632d2d69928fb0 | allow                      | 596059883112 |
| <input type="checkbox"/> | -    | sg-08a48ee48f6333df1  | frontend-asg-sg     | vpc-026632d2d69928fb0 | Allow                      | 596059883112 |
| <input type="checkbox"/> | -    | sg-0fd593737de6de05b  | backend-alb-sg      | vpc-026632d2d69928fb0 | allow                      | 596059883112 |
| <input type="checkbox"/> | -    | sg-0ff4fee6f31e26e385 | frontend-alb-sg     | vpc-026632d2d69928fb0 | Allow                      | 596059883112 |
| <input type="checkbox"/> | -    | sg-07269e306d1cd597b  | default             | vpc-026632d2d69928fb0 | default VPC security group | 596059883112 |

The inbound of frontend-alb is http traffic from internet.

The inbound of frontend-asg is http traffic from frontend-alb.

The inbound of backend-alb is http traffic from frontend-asg.

The inbound of backend-asg is http traffic from backend-alb.

The inbound of db-sg is MYSQL traffic at port 3306.

iv. The next was to create the database instance and database subnet group.

my-db

Summary

DB identifier

my-db

CPU

2.80%

Status

Available

Class

db.m5.large

Role

Instance

Current activity

0.00 sessions

Engine

MySQL Community

Region & AZ

eu-north-1a

Recommendations

Connectivity & security

Monitoring

Logs & events

Configuration

Zero-ETL integrations

Maintenance & backups

Data migrations

Tags

Recommendations

Connect using

Code snippets

Use when connecting through SDK, APIs, or third-party tools including agents.

CloudShell

Use for a quick access to AWS CLI that launches directly from the AWS Management Console.

Endpoints

Use when connecting through any IDE interface.

Internet access gateway

Disabled

IAM Authentication

Disabled

Programming language

MySQL (macOS)

Endpoint type

Instance endpoint

Connection steps

Follow the steps below to paste the code of each step in your tool and run the commands. The snippets dynamically reflect the authentication configuration.

1

mysql -h my-db.czggwgqayuae.eu-north-1.rds.amazonaws.com -P 3306 -u admin -p'<Enter\_DB\_Password>' --ssl-verify-server-cert --ssl-ca=/certs/global-bundle.pem mysql

Connected compute resources (0)

Connections to compute resources that were created automatically by RDS are shown here. Connections to compute resources that were created manually aren't shown.

Filter by compute resources

Resource identifier

Resource type

Availability Zone

VPC security group

Compute resource security group

Connected proxy

The subnet groups are and associated with the database instance.

mysubnetgroup

Subnet group details

VPC ID

vpc-026632d2d69928fb0

ARN

arn:aws:rds:eu-north-1:596059883112:subgrp:mysubnetgroup

Supported network types

IPv4

Description

This is my db subnet group.

Subnets (2)

| Availability zone | Subnet name    | Subnet ID                | CIDR block   |
|-------------------|----------------|--------------------------|--------------|
| eu-north-1b       | Private-Sub-04 | subnet-0a20f28624959cc32 | 10.0.80.0/20 |
| eu-north-1a       | Private-Sub-03 | subnet-06f4c86051946fe08 | 10.0.64.0/20 |

Tags (0)

Filter by Tag key

Tags

Value

You don't have any tags associated with this resource.

Manage tags

v. Now creating ALB and target group to be associated with it.

## 1. Creating frontend-alb and the frontend target group .

Introducing ALB target optimizer  
Target optimizer lets you enforce a maximum number of requests per target using an ALB-provided agent, improving success rates, latency, and efficiency. [Learn more](#)

frontend-alb

Details

Load balancer type

Application

Status

Active

VPC

vpc-026632d2d69928fb0

Load balancer IP address type

IPv4

Scheme

Internet-facing

Hosted zone

Z23TAZ6LKFMNIO

Availability Zones

subnet-09d3889f1303e58e1 eu-north-1b (eun1-az2)  
subnet-02622ff874c4a9747 eu-north-1a (eun1-az1)

Date created

February 5, 2026, 22:19 (UTC+05:30)

Load balancer ARN

arn:aws:elasticloadbalancing:eu-north-1:596059883112:loadbalancer/app/frontend-alb/66f72c1fb85ab0c2

DNS name

frontend-alb-1881685455.eu-north-1.elb.amazonaws.com (A Record)

Listeners and rules

Network mapping

Resource map

Security

Monitoring

Integrations

Attributes

Capacity

Tags

Listeners and rules (1)

Manage rules

Manage listener

Add listener

A listener checks for connection requests on its configured protocol and port. Traffic received by the listener is routed according to the default action and any additional rules.

Filter listeners

< 1 >

Protocol:Port

Default action

Rules

ARN

Security policy

Default SSL/TLS certificate

mTLS

Trust store

HTTP:80

Forward to target group  
frontend: 1 (100%)  
Target group stickiness: Off

1 rule

ARN

Not applicable

Not applicable

Not applicable

Not applicable

frontend

Actions

Details

arn:aws:elasticloadbalancing:eu-north-1:596059883112:targetgroup/frontend/5ec28a198390e26e

Target type

Instance

Protocol : Port

HTTP: 80

VPC

vpc-026632d2d69928fb0

IP address type

IPv4

Load balancer

frontend-alb

2

2 Healthy

0 Unhealthy

0 Unused

0 Initial

0 Draining

0 Anomalous

Distribution of targets by Availability Zone (AZ)

Select values in this table to see corresponding filters applied to the Registered targets table below.

Targets

Monitoring

Health checks

Attributes

Tags

Registered targets (2)

Anomaly mitigation: Not applicable

Deregister

Register targets

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

Filter targets

< 1 >

Instance ID

Name

Port

Zone

Health status

Health status details

Administrative o...

Override details

Launch...

Anomaly

I-041c4a1438f387006

-

80

eu-north-1a (e...

Healthy

-

No override

No override is curre...

February ...

Normal

I-0d01edc9ecd79f5ef

-

80

eu-north-1b (e...

Healthy

-

No override

No override is curre...

February ...

Normal

## 2. Creating backend-alb and the backend target group.

backend-alb

Actions

Details

Load balancer type

Application

Status

Active

VPC

vpc-026632d2d69928fb0

Load balancer IP address type

IPv4

Scheme

Internal

Hosted zone

Z23TAZ6LKFMNIO

Availability Zones

subnet-0a20f28624959cc32 eu-north-1b (eun1-az2)  
subnet-06f4c86051946fe08 eu-north-1a (eun1-az1)

Date created

February 5, 2026, 22:20 (UTC+05:30)

Load balancer ARN

arn:aws:elasticloadbalancing:eu-north-1:596059883112:loadbalancer/app/backend-alb/72581846dc5797ff

DNS name

internal-backend-alb-140462122.eu-north-1.elb.amazonaws.com (A Record)

Listeners and rules

Network mapping

Resource map

Security

Monitoring

Integrations

Attributes

Capacity

Tags

Listeners and rules (1)

Manage rules

Manage listener

Add listener

A listener checks for connection requests on its configured protocol and port. Traffic received by the listener is routed according to the default action and any additional rules.

Filter listeners

< 1 >

Protocol:Port

Default action

Rules

ARN

Security policy

Default SSL/TLS certificate

mTLS

Trust store

HTTP:80

Forward to target group  
backend: 1 (100%)  
Target group stickiness: Off

1 rule

ARN

Not applicable

Not applicable

Not applicable

Not applicable

backend

Actions

Details

am:aws:elasticloadbalancing:eu-north-1:596059883112:targetgroup/backend/b73c1a1b20426cc1

Target type

Instance

Protocol : Port

HTTP: 80

Protocol version

HTTP1

VPC

vpc-026632d2d69928fb0

IP address type

IPv4

Load balancer

backend-alb

2

2

0

0

0

Total targets

Healthy

Unhealthy

Unused

Initial

Draining

0 Anomalous

Distribution of targets by Availability Zone (AZ)

Select values in this table to see corresponding filters applied to the Registered targets table below.

Targets

Monitoring

Health checks

Attributes

Tags

Registered targets (2)

Anomaly mitigation: Not applicable

Deregister

Register targets

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

Filter targets

Instance ID

Name

Port

Zone

Health status

Health status details

Administrative o...

Override details

Launch...

Anomaly

I-0db85597b410f8894

-

80

eu-north-1b (e...

Healthy

-

No override

No override is curre...

February ...

Normal

I-054809a4feb70df72

-

80

eu-north-1a (e...

Healthy

-

No override

No override is curre...

February ...

Normal

vi. Now creating the launch template for frontend and the backend separately for the Auto Scaling group to use .

1. The frontend launch template is .

frontend-new (lt-052b0abb7eeacbbc)

Actions

Delete template

Launch template details

Launch template ID

lt-052b0abb7eeacbbc

Launch template name

frontend-new

Default version

1

Owner

am:aws:iam::596059883112:root

Details

Versions

Template tags

Launch template version details

Version

1 (Default)

Description

New Launch Template

Date created

2026-02-05T17:51:06.000Z

Created by

am:aws:iam::596059883112:root

Instance details

Storage

Resource tags

Network interfaces

Advanced details

AMI ID

ami-04233b5aece09244

Instance type

t3.small

Availability Zone

-

Availability Zone Id

-

Key pair name

Plyush

Security groups

-

Security group IDs

sg-08a48ee48f6333df1

2. The backend launch template is .

backend (lt-0d25b7ea21f96356d)

Actions

Delete template

Launch template details

Launch template ID

lt-0d25b7ea21f96356d

Launch template name

backend

Default version

1

Owner

am:aws:iam::596059883112:root

Details

Versions

Template tags

Launch template version details

Version

1 (Default)

Description

Launch template for backend

Date created

2026-02-05T17:06:41.000Z

Created by

am:aws:iam::596059883112:root

Instance details

Storage

Resource tags

Network interfaces

Advanced details

AMI ID

ami-04233b5aece09244

Instance type

t3.small

Availability Zone

-

Availability Zone Id

-

Key pair name

Plyush

Security groups

-

Security group IDs

sg-0a644300e0dcada7b

vii. Now creating auto scaling group and attaching them to the load balancers.  
The auto scaling group for frontend is :

frontend-new-asg

frontend-new-asg Capacity overview

Edit

arn:aws:autoscaling:eu-north-1:596059883112:autoScalingGroup:c5cce54b-6106-4505-bd5d-9b960d6a9878:autoScalingGroupName/frontend-new-asg

|                  |                |                             |        |
|------------------|----------------|-----------------------------|--------|
| Desired capacity | Scaling limits | Desired capacity type       | Status |
| 2                | 2 - 3          | Units (number of instances) | -      |

Date created  
Thu Feb 05 2026 23:22:24 GMT+0530 (India Standard Time)

Details

Integrations

Automatic scaling

Instance management

Instance refresh

Activity

Monitoring

Tags - moved

Launch template

Edit

Launch template

ami-04233b5a

Instance type

Owner

Version

Security groups

Security group IDs

Create time

Description

Storage (volumes)

Key pair name

Request Spot Instances

View details in the launch template console

Network

Edit

Availability Zones

Subnet ID

Availability Zone distribution

Instance type requirements

Edit

Health checks

Edit

Health check type

Health check grace period

Instance maintenance policy

Edit

Replacement behavior

Min healthy percentage

Max healthy percentage

Capacity Reservation preference

Edit

Preference

Capacity Reservation IDs

Resource Groups

Advanced configurations

Edit

Instance scale-in protection

Termination policies

Maximum instance lifetime

Service-linked role

Placement group

Suspended processes

Default cooldown

Default instance warmup

Auto Scaling group deletion protection

The auto scaling group for backend is :

backend-asg

backend-asg Capacity overview

Edit

arn:aws:autoscaling:eu-north-1:596059883112:autoScalingGroup:561f22f3-5a1b-4a31-9e0c-721c801c8a4d:autoScalingGroupName/backend-asg

|                  |                |                             |        |
|------------------|----------------|-----------------------------|--------|
| Desired capacity | Scaling limits | Desired capacity type       | Status |
| 2                | 2 - 3          | Units (number of instances) | -      |

Date created  
Thu Feb 05 2026 22:39:18 GMT+0530 (India Standard Time)

Details

Integrations

Automatic scaling

Instance management

Instance refresh

Activity

Monitoring

Tags - moved

Launch template

Edit

Launch template

ami-04233b5a

Instance type

Owner

Version

Security groups

Security group IDs

Create time

Description

Storage (volumes)

Key pair name

Request Spot Instances

View details in the launch template console

Network

Edit

Availability Zones

Subnet ID

Availability Zone distribution

Instance type requirements

Edit

Your Auto Scaling group adheres to the launch template for purchase option and instance type.

Health checks

Edit

|                   |                           |
|-------------------|---------------------------|
| Health check type | Health check grace period |
| EC2               | 300                       |

Instance maintenance policy

Edit

|                      |                        |                        |
|----------------------|------------------------|------------------------|
| Replacement behavior | Min healthy percentage | Max healthy percentage |
| No policy            | -                      | -                      |

Capacity Reservation preference

Edit

|            |                          |                 |
|------------|--------------------------|-----------------|
| Preference | Capacity Reservation IDs | Resource Groups |
| Default    | -                        | -               |

Advanced configurations

Edit

|                                        |                      |                           |                                                                                                        |
|----------------------------------------|----------------------|---------------------------|--------------------------------------------------------------------------------------------------------|
| Instance scale-in protection           | Termination policies | Maximum instance lifetime | Service-linked role                                                                                    |
| Not protected from scale in            | Default              | -                         | arn:aws:iam::596059883112:role/aws-service-role/autoscaling.amazonaws.com/AWSServiceRoleForAutoScaling |
| Placement group                        | Suspended processes  | Default cooldown          | Default instance warmup                                                                                |
| -                                      | -                    | 300                       | Disabled                                                                                               |
| Auto Scaling group deletion protection |                      |                           |                                                                                                        |
| None (default)                         |                      |                           |                                                                                                        |

Now the application is deployed successfully and it looks like this .

CloudKeeper

Basic Three-Tier Architecture Example

Message from Database:

- Hello World. This is Piyush Testing. (2026-02-07 05:31:12)
- Hello , Testing! (2026-02-05 17:54:50)
- Hello from the database! (2026-02-05 16:45:43)
- Welcome to our three-tier architecture demo (2026-02-05 16:45:43)
- This is a simple example showing frontend, backend, and database (2026-02-05 16:45:43)

Add New Message

New Message:

Save Message

**Q2. Create an instance in a private subnet (without assigning a public IP) with a NAT Gateway attached. SSH into it using a jump host and try accessing the internet. Take**

a screenshot. Then, create a new route table, attach an Internet Gateway (IGW) to it, and associate this route table with the private subnet where the instance was created. Now, try accessing the internet from the instance again. Is the internet reachable? Specify the reason for your findings and explain why it is reachable or not. Also, include screenshots to support your observations.

Ans :

I created the an instance in the private subnet with no public IP assigned to it .

Instance summary for i-Of3209307f1fcada1 (Q2) [Info](#)

Updated less than a minute ago

[Connect](#) [Instance state](#) [Actions](#)

|                                                                            |                                                                                                         |                                                                                                                                         |
|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Instance ID</b><br><a href="#">i-Of3209307f1fcada1</a>                  | <b>Public IPv4 address</b><br>-                                                                         | <b>Private IPv4 addresses</b><br><a href="#">10.0.42.42</a>                                                                             |
| <b>IPv6 address</b><br>-                                                   | <b>Instance state</b><br><a href="#">Running</a>                                                        | <b>Public DNS</b><br>-                                                                                                                  |
| <b>Hostname type</b><br>IP name: ip-10-0-42-42.eu-north-1.compute.internal | <b>Private IP DNS name (IPv4 only)</b><br><a href="#">ip-10-0-42-42.eu-north-1.compute.internal</a>     | <b>Elastic IP addresses</b><br>-                                                                                                        |
| <b>Answer private resource DNS name</b><br>-                               | <b>Instance type</b><br>t3.micro                                                                        | <b>AWS Compute Optimizer finding</b><br><a href="#">Opt-in to AWS Compute Optimizer for recommendations.</a> <a href="#">Learn more</a> |
| <b>Auto-assigned IP address</b><br>-                                       | <b>VPC ID</b><br><a href="#">vpc-026632d2d69928fb0</a>                                                  | <b>Auto Scaling Group name</b><br>-                                                                                                     |
| <b>IAM Role</b><br>-                                                       | <b>Subnet ID</b><br><a href="#">subnet-07d1352941da01430</a>                                            | <b>Managed</b><br>false                                                                                                                 |
| <b>IMDSv2</b><br>Required                                                  | <b>Instance ARN</b><br><a href="#">arn:aws:ec2:eu-north-1:596059883112:instance/i-Of3209307f1fcada1</a> |                                                                                                                                         |
| <b>Operator</b><br>-                                                       |                                                                                                         |                                                                                                                                         |

[Details](#) [Status and alarms](#) [Monitoring](#) [Security](#) [Networking](#) [Storage](#) [Tags](#)

▼ **Instance details** [Info](#)

|                                                                                    |                                                               |                                                                                     |
|------------------------------------------------------------------------------------|---------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>AMI ID</b><br><a href="#">ami-04233b5aecc09244</a>                              | <b>Monitoring</b><br>disabled                                 | <b>Platform details</b><br><a href="#">Linux/UNIX</a>                               |
| <b>AMI name</b><br><a href="#">al2023-ami-2023.10.20260120.4-kernel-6.1-x86_64</a> | <b>Allowed image</b><br>-                                     | <b>Termination protection</b><br>Disabled                                           |
| <b>Stop protection</b><br>Enabled                                                  | <b>Launch time</b><br><a href="#">2023-10-20 10:20:26 UTC</a> | <b>AMI location</b><br><a href="#">eu-north-1:596059883112:ami-04233b5aecc09244</a> |

Now creating the jump host with public IP enabled to it , so that I can ssh into my private ec2 instance in the private subnet.

Instance summary for i-08971c0bfa600c61e (JumpHost) [Info](#)

Updated less than a minute ago

[Connect](#) [Instance state](#) [Actions](#)

|                                                                              |                                                                                                         |                                                                                                                                         |
|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Instance ID</b><br><a href="#">i-08971c0bfa600c61e</a>                    | <b>Public IPv4 address</b><br><a href="#">51.21.194.183</a> <a href="#">open address</a>                | <b>Private IPv4 addresses</b><br><a href="#">10.0.22.211</a>                                                                            |
| <b>IPv6 address</b><br>-                                                     | <b>Instance state</b><br><a href="#">Running</a>                                                        | <b>Public DNS</b><br>-                                                                                                                  |
| <b>Hostname type</b><br>IP name: ip-10-0-22-211.eu-north-1.compute.internal  | <b>Private IP DNS name (IPv4 only)</b><br><a href="#">ip-10-0-22-211.eu-north-1.compute.internal</a>    | <b>Elastic IP addresses</b><br>-                                                                                                        |
| <b>Answer private resource DNS name</b><br>-                                 | <b>Instance type</b><br>t3.micro                                                                        | <b>AWS Compute Optimizer finding</b><br><a href="#">Opt-in to AWS Compute Optimizer for recommendations.</a> <a href="#">Learn more</a> |
| <b>Auto-assigned IP address</b><br><a href="#">51.21.194.183</a> [Public IP] | <b>VPC ID</b><br><a href="#">vpc-026632d2d69928fb0 (3TierVPC)</a>                                       | <b>Auto Scaling Group name</b><br>-                                                                                                     |
| <b>IAM Role</b><br>-                                                         | <b>Subnet ID</b><br><a href="#">subnet-09d3889f1303e58e1 (Public-Sub-02)</a>                            | <b>Managed</b><br>false                                                                                                                 |
| <b>IMDSv2</b><br>Required                                                    | <b>Instance ARN</b><br><a href="#">arn:aws:ec2:eu-north-1:596059883112:instance/i-08971c0bfa600c61e</a> |                                                                                                                                         |
| <b>Operator</b><br>-                                                         |                                                                                                         |                                                                                                                                         |

[Details](#) [Status and alarms](#) [Monitoring](#) [Security](#) [Networking](#) [Storage](#) [Tags](#)

▼ **Instance details** [Info](#)

|                                                                                    |                                                               |                                                                                     |
|------------------------------------------------------------------------------------|---------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>AMI ID</b><br><a href="#">ami-04233b5aecc09244</a>                              | <b>Monitoring</b><br>disabled                                 | <b>Platform details</b><br><a href="#">Linux/UNIX</a>                               |
| <b>AMI name</b><br><a href="#">al2023-ami-2023.10.20260120.4-kernel-6.1-x86_64</a> | <b>Allowed image</b><br>-                                     | <b>Termination protection</b><br>Disabled                                           |
| <b>Stop protection</b><br>Enabled                                                  | <b>Launch time</b><br><a href="#">2023-10-20 10:20:26 UTC</a> | <b>AMI location</b><br><a href="#">eu-north-1:596059883112:ami-04233b5aecc09244</a> |



Now I SSH into my jump host and then copied the Private key pair in my ec2 instance so that i can SSH into my private ec2 instance.

```
ec2-user@ip-10-0-42-42:~  
piyush@CK-piyush:~$ ssh -i ~/Downloads/Piyush.pem ec2-user@51.21.194.183  
#_ _  
~\_\_####_ Amazon Linux 2023  
~~ \_\_#####\  
~~ \_\_###|  
~~ \_\_#/____ https://aws.amazon.com/linux/amazon-linux-2023  
~~ V~' '->  
~~~ /  
~~.-._ -/_  
_/_/_/_/  
_/m/' '  
  
Last login: Sat Feb 7 05:53:12 2026 from 103.83.128.50  
[ec2-user@ip-10-0-22-211 ~]$ ls -al  
total 20  
drwx----- . 3 ec2-user ec2-user 113 Feb 7 06:01  
drwxr-xr-x .. 3 root root 22 Feb 7 05:40  
-rw----- .bash_history 1 ec2-user ec2-user 137 Feb 7 05:58  
-rw-r--r-- .bash_logout 1 ec2-user ec2-user 18 Jan 28 2023  
-rw-r--r-- .bash_profile 1 ec2-user ec2-user 141 Jan 28 2023  
-rw-r--r-- .bashrc 1 ec2-user ec2-user 492 Jan 28 2023  
drwx----- .ssh 2 ec2-user ec2-user 48 Feb 7 05:53  
-r----- Piyush.pem 1 ec2-user ec2-user 1678 Feb 7 06:01  
[ec2-user@ip-10-0-22-211 ~]$ ssh -i "Piyush.pem" ec2-user@10.0.42.42
```

Then using the key and jump host I successfully SSH into private instance which have a NAT gateway attached to it in public subnet , so the internet connectivity should work. I used `ping google.com` to check the internet connectivity.

```
ec2-user@ip-10-0-42-42:~  
[ec2-user@ip-10-0-42-42 ~]$ ping google.com  
PING google.com (142.251.38.110) 56(84) bytes of data.  
64 bytes from lcarna-ac-in-f14.1e100.net (142.251.38.110): icmp_seq=1 ttl=118 time=3.91 ms  
64 bytes from lcarna-ac-in-f14.1e100.net (142.251.38.110): icmp_seq=2 ttl=118 time=3.73 ms  
64 bytes from lcarna-ac-in-f14.1e100.net (142.251.38.110): icmp_seq=3 ttl=118 time=3.71 ms  
64 bytes from lcarna-ac-in-f14.1e100.net (142.251.38.110): icmp_seq=4 ttl=118 time=3.81 ms
```

The connection succeeds as expected , because the NAT gateway is in public subnet , for successful internet connection the NAT gateway should be in public subnet and attached to IGW which allows the internet connectivity for the private ec2 instance.

Now creating the route table and attaching it to private subnet to it. Then adding the route for IGW in the route table, the ec2- instance connection fails because for successful internet connectivity a device needs a public IP address if it is attached to IGW directly . Since our ec2-instance is private it has no public IP assigned to it and hence the connection fails.

rtb-0eb7fbbf61d5ee3b2 / Q2RT Actions

**Details** Info  
**Route table ID**  
rtb-0eb7fbbf61d5ee3b2  
**VPC**  
vpc-026632d2d69928fbo | 3TierVPC

**Main**  
No  
**Owner ID**  
596059883112

**Explicit subnet associations**  
subnet-07d1352941da01430 / Private-Sub-01

**Edge associations**  
-

[Routes](#) | [Subnet associations](#) | [Edge associations](#) | [Route propagation](#) | [Tags](#)

**Routes (2)** Both Edit routes

Filter routes

| Destination | Target                | Status | Propagated | Route Origin       |
|-------------|-----------------------|--------|------------|--------------------|
| 0.0.0.0/0   | igw-0b8b72c09743c8a95 | Active | No         | Create Route       |
| 10.0.0.0/16 | local                 | Active | No         | Create Route Table |

ec2-user@ip-10-0-42-42:~

ping google.com

PING google.com (142.251.38.110) 56(84) bytes of data.

^?^C

--- google.com ping statistics ---

9 packets transmitted, 0 received, 100% packet loss, time 8353ms

[ec2-user@ip-10-0-42-42 ~]\$

Now the following observations were made :

We have Private Instance + NAT gateway attached to it , so the connection succeeded because the translation of private IP to public IP by NAT gateway allowed the internet connection.

The second scenario was that We added a route to IGW for private instance with no NAT so the connection failed because the ec2 instance did not had any public IP so the IGW dropped the packets without routing it to the internet .

So for a successful and proper internet connection we need 1. Public IP (Must have) 2. Route Tables configured for routing 3. Internet Gateway.

**Q3. Explain the importance of the Internet Gateway (IGW), Public IP / Elastic IP, and NAT Gateway. Can we access the internet with only a route to an IGW and no public IP? If yes, why? If no, why?**

**Ans :**

**Internet Gateway :** It is an important device which allows inbound and outbound internet access to all the devices / subnets within a VPC . Without Internet Gateway the internet connection is not possible even if you have configured the route tables and have a public IP with you . It allows the traffic to flow from and to a VPC.

**Public IP :** It is the most crucial and important component for the instances to connect to the internet. Without Public IP not internet connection is possible. This Public IP is assigned to instances when they are launched.

**Elastic IP:** It is the same as Public IP , does the same job as the public IP but the only difference is that it is static , it does not change when you stop and reboot the instance whereas the public IP gets changed. AWS charges for it , if it is not associated with any instance.

**No , We cannot access the internet with only a route to an IGW and No public IP :** This is because public IP plays an crucial part to connect successfully to the internet , it is the address by which the instance is identified on the internet . Without it no-one can recognize the instance and the connection fails .

The internet connection works like this if the instances are in public subnet with public IP attached :

**EC2 Instance -> IGW -> Internet**

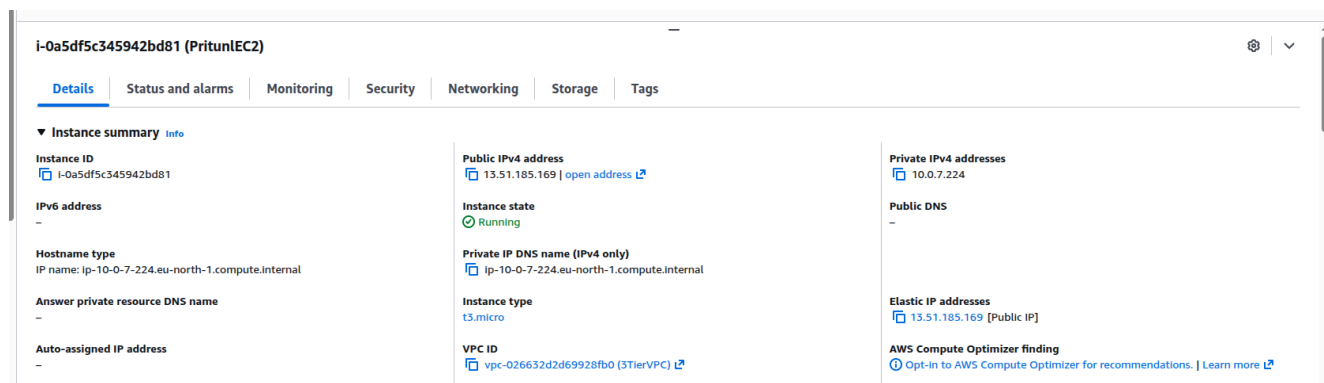
The internet connection works like this if the instances are in private subnets and have NAT gateway attached to them in public subnets .

**EC2 Instance - > NAT (in public subnet) -> IGW -> Internet**

## Q4. Set up a VPN (Pritunl) and then try accessing the internal network of your VPC and its private resources.

**Ans :**

To create a VPN ( Pritunl) , I first created an ec2 instance with auto public IP disabled and then associated elastic IP because this will create a VPN tunnel to connect to private resources in the VPC .



Now ssh into it to first configure and install the pritunl on the instance created . The instance is created in the public subnet to make access easier and configure it for the VPN connection .

Now using Pritunl to setup the VPN connection now

Adding the server and vpc route for creating the VPN tunnel to access private resources of my VPC.

Attached the pritunl server to my vpc and added the route from here to the vpc cidr.

**pritonl** Dashboard Users Servers Upgrade to Enterprise! Logs Settings Logout

## Servers

[Add Server](#) [Add Route](#) [Attach Organization](#)

Successfully saved server settings. ×

Successfully added route. ×

Successfully attached organization. ×

**Server** my-vpc-vpn Settings Restart Server Stop Server Delete Server

**Status** Online Server Output Bandwidth Graphs

**Uptime** 0d 0h 0m 8s

**Users** 0/1 users online

**Devices** 0 devices online

**Network** 10.99.0.0/24

**Port** 1194/udp

**Multiple Devices** Disabled

```

1 [cobalt-3997] 2026-02-07 10:04:46 Note: Treating option '--ncp-ciphers' as '--data-ciphers' (renamed in OpenVPN 2.5).
2 [cobalt-3997] 2026-02-07 10:04:46 OpenVPN 2.6.14 x86_64-pc-linux-gnu [SSL (OpenSSL)] [LZO] [LZ4] [EPOLL] [PKCS11] [MH/PTKINFO] [
3 [cobalt-3997] 2026-02-07 10:04:46 library versions: OpenSSL 3.0.13 30 Jan 2024, LZO 2.10
4 [cobalt-3997] 2026-02-07 10:04:46 DCO version: N/A
5 [cobalt-3997] 2026-02-07 10:04:46 TUN/TAP device tun0 opened
6 [cobalt-3997] 2026-02-07 10:04:46 net_iface_mtu_set: mtu 1500 for tun0
7 [cobalt-3997] 2026-02-07 10:04:46 net_iface_up: set tun0 up
8 [cobalt-3997] 2026-02-07 10:04:46 net_addr_v4_add: 10.99.0.1/24 dev tun0
9 [cobalt-3997] 2026-02-07 10:04:46 setsockopt(TPVG_V6ONLY=0)
10 [cobalt-3997] 2026-02-07 10:04:46 UDPv6 link local (bound): [AF_INET6][undef]:1194
11 [cobalt-3997] 2026-02-07 10:04:46 UDPv6 link remote: [AF_UNSPEC]
12 [cobalt-3997] 2026-02-07 10:04:46 Initialization Sequence Completed
13 [cobalt-3997] 2026-02-07 10:04:48 COM> SUCCESS: bytecount interval changed
14

```

**0.0.0.0/0** Remove Route

**10.99.0.0/24** Virtual Network Remove Route

**10.0.0.0/16** Remove Route

**8.8.8.8/32 DNS Server** Remove Route

**my-vpc-org** Detach Organization

Now I downloaded the user and connected it from laptop , now i am successfully inside the vpc.

```

piyush@CK-piyush: ~/Downloads$ sudo openvpn --config my-vpc-org_my-vpc-user_my-vpc-vpn.ovpn
2026-02-07 15:41:29 OpenVPN 2.6.14 x86_64-pc-linux-gnu [SSL (OpenSSL)] [LZO] [LZ4] [EPOLL] [PKCS11] [MH/PTKINFO] [AEAD] [DCO]
2026-02-07 15:41:29 library versions: OpenSSL 3.0.13 30 Jan 2024, LZO 2.10
2026-02-07 15:41:29 DCO version: N/A
2026-02-07 15:41:29 TCP/UDP: Preserving recently used remote address: [AF_INET]13.50.154.133:1194
2026-02-07 15:41:29 UDPv4 link local: (not bound)
2026-02-07 15:41:29 UDPv4 link remote: [AF_INET]13.50.154.133:1194
2026-02-07 15:41:29 VERIFY OK: depth=1, O=69870c3cc0820a8f708d14ae, CN=69870c3cc0820a8f708d14af
2026-02-07 15:41:29 NOTE: --mute triggered...
2026-02-07 15:41:29 6 variation(s) on previous 3 message(s) suppressed by --mute
2026-02-07 15:41:29 [69870c3cc0820a8f708d14bc] Peer Connection Initiated with [AF_INET]13.50.154.133:1194
2026-02-07 15:41:30 TUN/TAP device tun0 opened
2026-02-07 15:41:30 net_iface_mtu_set: mtu 1500 for tun0
2026-02-07 15:41:30 net_iface_up: set tun0 up
2026-02-07 15:41:30 net_addr_v4_add: 10.99.0.2/24 dev tun0
2026-02-07 15:41:30 Initialization Sequence Completed
2026-02-07 15:41:30 Data Channel: cipher 'AES-128-GCM', peer-id: 0, compression: 'stub'
2026-02-07 15:41:30 NOTE: --mute triggered...
2026-02-07 15:43:27 2 variation(s) on previous 3 message(s) suppressed by --mute
2026-02-07 15:43:27 AEAD Decrypt error: bad packet ID (may be a replay): [ #996 ] -- see the man page entry for --no-replay and --replay-window for more info or s
ute-replay-warnings
2026-02-07 15:43:59 NOTE: --mute triggered...

```

I can check the route to my vpc

```

piyush@CK-piyush: ~/Downloads$ ip route | grep 10.0.0.0
10.0.0.0/16 via 10.99.0.1 dev tun0
piyush@CK-piyush: ~/Downloads$

```

Now trying to accessing the private resources in the vpc by using their private IP and pinging them to see if the connection is working and the vpc tunnel is working correctly.

Now I have my RDS MYSQL running in my private subnets , so by using the command `nc -vz my-db.czggwgqayuae.eu-north-1.rds.amazonaws.com 3306` , I see that the

connection is successful and the VPN tunnel works fine.

I am able to access private resources from it.

```
piyush@CK-piyush:~/Downloads$ nc -vz my-db.czggwgqayuoe.eu-north-1.rds.amazonaws.com 3306
Connection to my-db.czggwgqayuoe.eu-north-1.rds.amazonaws.com (10.0.71.255) 3306 port [tcp/mysql] succeeded!
piyush@CK-piyush:~/Downloads$
```

The Pritunl UI also gives logs to see the connection details.

The screenshot displays the Pritunl web interface for a server named 'my-vpc-vpn'. The interface includes a sidebar with navigation links for Status, Uptime, Users, Devices, Network, Port, and Multiple Devices. The main content area shows the 'Server Output' tab, which contains a log of connection events. The logs show a successful connection from 10.0.0.0/0 to 10.99.0.0/24, followed by a client kill command and a successful authentication. The interface also features buttons for 'Settings', 'Restart Server', 'Stop Server', and 'Delete Server'. Below the logs, there are sections for 'Routes' and 'DNS Servers', each with a 'Remove Route' button. At the bottom, there is a 'Detach Organization' button.

**Q5. A fleet of 4 EC2 instances is running behind an Application Load Balancer. You need to roll out a security patch by creating a new version of the Launch Template used by the Auto Scaling group. During the update process, at least 80% of the fleet's serving capacity must remain available at all times. How can you achieve this? Implement this update in AWS by configuring the Auto Scaling group to use the new Launch Template version and perform the rollout in a way that ensures at least 80% of the instances remain healthy and in service behind the Application Load Balancer throughout the update. Explain the configuration choices you make and verify availability during the process.**

**Ans:**

This is my current ASG which has a fleet of 4 ec2 instances running behind an application load balancer with the launch template version 1. The template has a simple html script to

display the message that current instances are running on version 1.

EC2 > Auto Scaling groups

Auto Scaling groups (1/1) [Info](#) Last updated less than a minute ago [Launch configurations](#) [Launch templates](#) [Actions](#) [Create Auto Scaling group](#)

Search your Auto Scaling groups

| Name       | Launch template/configuration | Instances | Status | Desired capacity | Min | Max | Availability Zones   | Creation time                                           |
|------------|-------------------------------|-----------|--------|------------------|-----|-----|----------------------|---------------------------------------------------------|
| my-web-asg | Web-App   Version Default     | 4         | -      | 4                | 4   | 6   | 2 Availability Zones | Sun Feb 08 2026 11:06:34 GMT+0530 (India Standard Time) |

Auto Scaling group: my-web-asg

[Details](#) [Integrations](#) [Automatic scaling](#) [Instance management](#) [Instance refresh](#) [Activity](#) [Monitoring](#) [Tags - moved](#)

**my-web-asg Capacity overview** [Edit](#)

arn:aws:autoscaling:eu-north-1:596059883112:autoScalingGroup:22ef1558-7722-4c0f-bdda-aceb223a2251:autoScalingGroupName/my-web-asg

| Desired capacity | Scaling limits | Desired capacity type       | Status |
|------------------|----------------|-----------------------------|--------|
| 4                | 4 - 6          | Units (number of Instances) | -      |

Date created  
Sun Feb 08 2026 11:06:34 GMT+0530 (India Standard Time)

**Launch template** [Edit](#)

| Launch template                 | AMI ID                | Instance type | Owner                         |
|---------------------------------|-----------------------|---------------|-------------------------------|
| lt-0ab22f868ea005ba6<br>Web-App | ami-04233b5aeece09244 | t3.micro      | am:aws:iam::596059883112:root |

| Version | Security groups | Security group IDs   | Create time                                             |
|---------|-----------------|----------------------|---------------------------------------------------------|
| Default | -               | sg-0482968ab9002823a | Sun Feb 08 2026 11:00:49 GMT+0530 (India Standard Time) |

my-web-alb-888779225.eu-north-1.elb.amazonaws.com

Hello This is Piyush - Version 1

All the instances and the application running currently are healthy.

my-web-tg [Actions](#)

**Details**

arn:aws:elasticloadbalancing:eu-north-1:596059883112:targetgroup/my-web-tg/6592585540022801

| Target type | Protocol : Port | Protocol version | VPC                   |
|-------------|-----------------|------------------|-----------------------|
| Instance    | HTTP: 80        | HTTP1            | vpc-026632d2d69928fb0 |

| IP address type | Load balancer |
|-----------------|---------------|
| IPv4            | my-web-alb    |

| Total targets | Healthy | Unhealthy | Unused | Initial | Draining |
|---------------|---------|-----------|--------|---------|----------|
| 4             | 4       | 0         | 0      | 0       | 0        |

0 Anomalous

**Distribution of targets by Availability Zone (AZ)**  
Select values in this table to see corresponding filters applied to the Registered targets table below.

[Targets](#) [Monitoring](#) [Health checks](#) [Attributes](#) [Tags](#)

**Registered targets (4)** [Info](#) [Anomaly mitigation: Not applicable](#) [Deregister](#) [Register targets](#)

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

Filter targets

| Instance ID         | Name | Port | Zone               | Health status | Health status details | Administrative override | Override details                       |
|---------------------|------|------|--------------------|---------------|-----------------------|-------------------------|----------------------------------------|
| i-056cb432c091df870 | -    | 80   | eu-north-1a (e...) | Healthy       | -                     | No override             | No override is currently active on ... |
| i-03829d7aade8b8529 | -    | 80   | eu-north-1a (e...) | Healthy       | -                     | No override             | No override is currently active on ... |
| i-00897f8e77a4a264e | -    | 80   | eu-north-1b (e...) | Healthy       | -                     | No override             | No override is currently active on ... |

Now rolling out the update patch .

So first modified the launch template to introduce the changes required by modifying user data script in my case.



**Launch Templates (1/1)** [Info](#)

Search

| Launch Template ID   | Launch Template Name | Default Version | Latest Version | Create Time              | Created By                     | Managed | Operator |
|----------------------|----------------------|-----------------|----------------|--------------------------|--------------------------------|---------|----------|
| lt-0ab22f868ea005ba6 | Web-App              | 1               | 2              | 2026-02-08T05:30:49.000Z | arn:aws:iam::596059883112:root | false   | -        |

**Web-App (lt-0ab22f868ea005ba6)**

**Launch template details**

Launch template ID: lt-0ab22f868ea005ba6  
 Launch template name: Web-App  
 Default version: 1  
 Owner: arn:aws:iam::596059883112:root

**Launch template version details**

Version: 2  
 Description: Security Patch V2  
 Date created: 2026-02-08T06:01:27.000Z  
 Created by: arn:aws:iam::596059883112:root

**Instance details**

AMI ID: ami-04233b5aeece09244  
 Instance type: t3.micro  
 Availability Zone: -  
 Availability Zone Id: -  
 Key pair name: Piyush  
 Security groups: -  
 Security group IDs: sg-0482968ab9002823a

Configured the instance refresh in ASG to maintain 80% of the fleet capacity while rolling out the update . AWS will maintain 4 instances because  $.80 * 4 = 3.2$  , AWS rounds up to 4 instances , meaning 4 instances should stay healthy at any given time.

Instance refresh started successfully.

Sun Feb 08 2026 11:06:34 GMT+0530 (India Standard Time)

Details | Integrations | Automatic scaling | Instance management | **Instance refresh** | Activity | Monitoring | Tags - moved

**Active instance refresh** [Info](#)

Instance refresh ID: 6186a566-687a-4367-aa9e-53b6cfb48ac5  
 Instance refresh status: Pending  
 Start time: -  
 CloudWatch alarm: Enabled

Minimum healthy percentage: 80%  
 Instance warmup: 180 seconds  
 Bake time: -

Skip matching: Enabled  
 Scale-in protected instances: Ignore  
 Checkpoints: -

Desired configuration: View  
 Auto rollback: Disabled  
 Standby instances: Ignore

**Instance refresh history (1)** [Info](#)

Filter instance refresh history

| Instance refresh ID                  | Instance refresh status | Rollback status | Status reason | Percentage completed | Instances to update | Start time | End time |
|--------------------------------------|-------------------------|-----------------|---------------|----------------------|---------------------|------------|----------|
| 6186a566-687a-4367-aa9e-53b6cfb48ac5 | Pending                 | -               | -             | -                    | -                   | -          | -        |

Feb 8 11:38

VPC | eu-north-1 | Instances | EC2 | eu-north-1 | Launch templates | EC2 | Load balancer details | Auto Scaling group details | my-web-alb-888779225

my-web-alb-888779225.eu-north-1.elb.amazonaws.com

**Hello This is Piyush - Version 2 (Patched Applied) This is more secure version.**

Configurations choices made were as follows :

1. First Modified the instance refresh configuration in ASG to 80% of the fleet , so that minimum 4 instances stay running before creating updating the rollout. We need to maintain at least 4 instances according to the 80% requirement producing no downtime.
2. Max Capacity > Desired Capacity because , the instances should be 4 before creating new one and then delete the old one to introduce the changes . So Max capacity should be greater than 4.

3. Instance warmup ensures that new version i.e version 2 is fully ready first and then kill the old version to take its place.